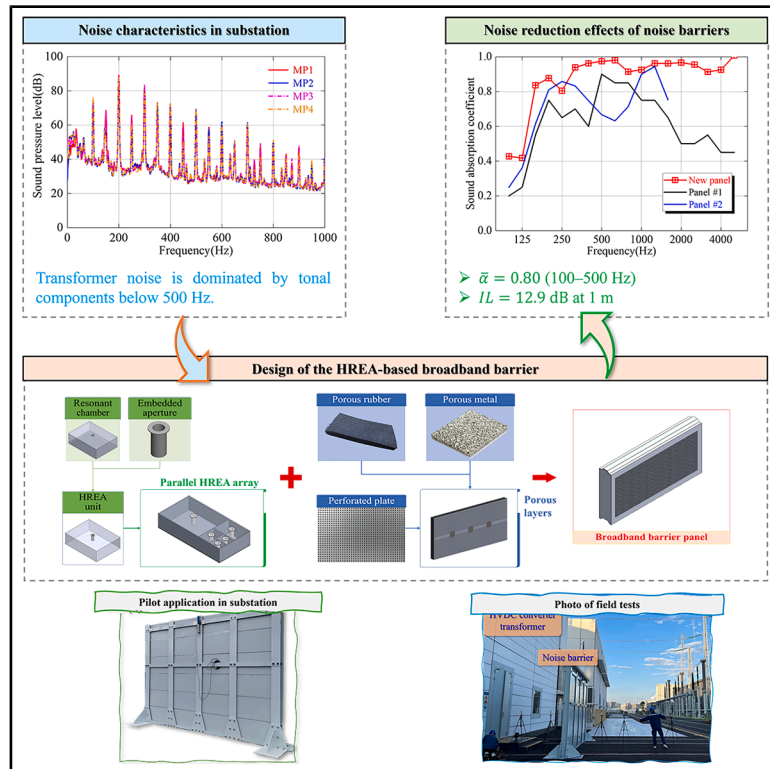


Design of a broadband noise reduction barrier utilizing acoustic metamaterial with application to a substation

Graphical abstract



Authors

Shenghui Xu, Li Li, Qi Wang, Wei Meng, Xing Du, Xinbiao Xiao, Zefeng Wen

Correspondence

duxing@swjtu.edu.cn

In brief

Applied sciences; devices; materials application

Highlights

- *In situ* tests identified transformer noise dominated by tonal components below 500 Hz
- Parallel HREA coupling generated strong absorption peaks at 100 Hz and its harmonics
- The hybrid barrier achieved an average absorption of 0.80 over 100–500 Hz
- Optimization enabled further gains or thickness reduction to 100 mm



Article

Design of a broadband noise reduction barrier utilizing acoustic metamaterial with application to a substation

Shenghui Xu,^{1,5,6} Li Li,^{2,6} Qi Wang,^{3,6} Wei Meng,^{2,6} Xing Du,^{4,5,6,7,8,*} Xinbiao Xiao,^{1,5,6} and Zefeng Wen^{1,5,6}¹State Key Laboratory of Rail Transit Vehicle System, Southwest Jiaotong University, Chengdu 610031, China²Electric Power Research Institute, Guangdong Power Grid Co., Ltd., Guangzhou 510030, China³CRRC Technology Innovation Co., Ltd., Beijing 100000, China⁴School of Electrical Engineering, Southwest Jiaotong University, Chengdu 610031, China⁵Present address: 111, Section 1, North Second Ring Road, Chengdu, Sichuan, China⁶These authors contributed equally⁷Senior author⁸Lead contact*Correspondence: duxing@swjtu.edu.cn<https://doi.org/10.1016/j.isci.2026.115918>

SUMMARY

Power transformer noise in substations is characterized by strong mid-low frequency components below 500 Hz, which are difficult to control using conventional noise barriers. To address this issue, *in situ* measurements were conducted to identify the acoustic features of transformer noise, and a broadband noise reduction barrier based on Helmholtz resonant absorption was developed. The barrier incorporates parallel Helmholtz resonators with embedded apertures and porous materials to enhance low-frequency energy dissipation and broaden the absorption bandwidth. Reverberation chamber tests and on-site measurements demonstrate that the proposed barrier achieves an average absorption coefficient of 0.80 within the 100–500 Hz in 1/3 octave bands and reduces the outdoor sound pressure level by 12.9 dB at 1 m outside the barrier, outperforming conventional absorbing barriers. The design also allows further performance improvement or thickness reduction through internal parameter optimization, indicating strong potential for practical mid-low frequency noise mitigation in substations.

INTRODUCTION

With the rapid growth of electricity demand, substations have increased in number and are being constructed increasingly close to residential areas, which have intensified concerns about environmental noise pollution.^{1,2} The dominant noise sources in substations, such as transformers, reactors, and capacitors, radiate mainly in the mid-low frequency range below 1,000 Hz, with pronounced tonal components at 100 Hz and its harmonics.^{3–5} Low-frequency noise (LFN) is particularly disturbing to humans and is associated with various adverse physiological and psychological effects even at moderate sound pressure levels.^{6–10} Therefore, efficient control of substation LFN has become an important and urgent engineering task. Although noise barriers are widely applied due to their simplicity and cost-effectiveness,¹¹ conventional barriers mainly rely on sound reflection and diffraction and show limited capability in dissipating acoustic energy in the low-frequency range. Field measurements indicate that existing substation barriers mainly alter sound propagation paths but exhibit insufficient absorption of transformer noise, as illustrated in Figure 1.¹²

In recent years, acoustic metamaterials and acoustic metastructures (AMs) have provided new possibilities for compact and efficient LFN control owing to their subwavelength resonance and impedance-tailoring capabilities.^{13–16} Especially, metastructures based on bending channels, labyrinthine geometries, and hybrid resonant mechanisms have shown strong potential for broadband sound absorption in thin configurations. Representative studies have reported broadband absorption using spatial bending metasurfaces¹⁷ and efficient inverse design strategies based on deep neural networks,¹⁸ indicating the rapid development of compact, high-performance, and intelligently designed AMs.

Meanwhile, increasing efforts have been devoted to extending AMs from laboratory-scale units to practical noise control devices such as noise barriers and acoustic enclosures. A composite resonant sound-absorbing barrier was developed to overcome the limitations of conventional noise barriers in the low-frequency band.¹⁹ Modular and inverse-designed AMs have been applied to ventilated noise barriers, demonstrating improved broadband sound insulation in engineering scenarios.²⁰ In addition, ultra-thin metasurface panels have been integrated into acoustic enclosures and verified experimentally to enhance



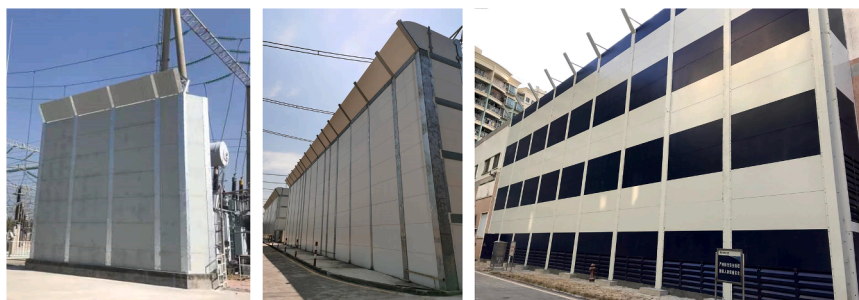


Figure 1. Existing noise barriers installed in substations

LFN reduction performance.²¹ These studies confirm the feasibility of employing AM concepts in large-scale noise barriers and enclosure systems.

Among various AM configurations, HR-based units and arrays remain one of the most fundamental and effective approaches for LFN mitigation due to their strong resonance characteristics and high design flexibility. Recent theoretical advances have further clarified that broadband absorption can be achieved through the coupling of multiple resonant modes and appropriate regulation of radiative and intrinsic losses. Fan optimized near-rail barriers for inter-city railways by utilizing HRs and analyzed the effects on noise at major frequency bands.²² In particular, the concept of quality-factor-weighted mode density has been proposed to explain and guide ultrabroadband absorption in HR-based metamaterial systems,²³ highlighting the significant potential of HREA-type architectures for compact broadband acoustic control.

Motivated by these advances, this study focuses on the transformer noise control in a typical substation. On-site tests were first conducted to characterize the spectral features of transformer noise and determine the dominant low-frequency targets. Based on the Helmholtz resonance mechanism, HREA units were designed to effectively suppress noise at 100 Hz and its harmonics. By combining HRs in parallel and utilizing the coherent coupling effect between HR units, an AM, with multiple absorption peaks at 100 Hz and its harmonics, was constructed to enhance the low-frequency absorption performance, and absorption performance was evaluated through simulations and tests. The proposed AMs were integrated with porous materials into the new barrier panel to enhance its absorption performance in the mid-low frequency band. Further optimization of the new barrier panel can be achieved by adjusting its internal components to improve its sound absorption performance and the NRE, or reduce its thickness while maintaining the original NRE. The design of the noise barrier followed the universal design principle, and both the sound absorption performance and the actual NREs were experimentally examined, aiming to provide a feasible and efficient solution for LFN control in substation applications.

Field test

To investigate the noise characteristics of power transformers, a field test was conducted on a typical transformer used in China. Through analyzing the test data, noise levels and spectral features of transformer noise were obtained. The photo of the transformer tested in the ± 800 kV ZTD station is shown in Figure 2A. Following the DL/T 1327–2014 standard, acoustic measurement points (MPs)

were positioned at 1-m horizontal intervals around the transformer, at a height of 1.5 m. The schematic of the MPs arrangement is shown in Figure 2B, with red dots indicating the locations of the MPs.

Noise level

The noise level at 1 m from the transformer was first analyzed. Table 1 presents the Z-weighted SPLs, which considered noise in the low frequency band (20–500 Hz), mid-low frequency band (20–1,000 Hz), and broadband (20–6,300 Hz), respectively.

From the data in Table 1, it can be observed that the Z-weighted SPLs for the low frequency noise and mid-low frequency noise differ by less than 0.1 dB from the SPLs that include the broadband noise. This indicates that transformer noise is primarily concentrated in the low frequency band. Therefore, for the subsequent design of noise barriers, it is essential to target and control noise in this frequency band to effectively reduce the SPLs of transformer noise. In addition, according to the existing test data, for the transformer in the ZTD station, the noise levels from different directions were nearly identical, with the SPLs differed by less than 1.5 dB.

Spectral feature

To identify significant frequencies of the transformer noise, the spectral features of the noise analyzed. Figure 3 shows the narrowband spectrum curves for the transformer noise in the ZTD station at several MPs.

As shown in Figure 3, the noise from the converter transformer in the ZTD station exhibited several SPL peaks at 50 Hz and its harmonics. The SPL values at even multiples of 50 Hz were slightly higher than those at adjacent odd multiples. The SPLs reached their maximum at 200 Hz, and the SPL peaks gradually decreased as the frequency increased.

Design

Sound absorption mechanism of HRs

The absorption coefficient (α) of acoustic materials or structures can be determined based on the normal acoustic impedance (Z). The relationship between α and Z can be described by²⁴

$$\alpha = 1 - \left| \frac{Z - Z_0}{Z + Z_0} \right|^2, \quad (\text{Equation 1})$$

where $Z_0 = \rho_0 c_0$ is the characteristic impedance of air. Here, $\rho_0 = 1.29 \text{ kg/m}^3$ and $c_0 = 343 \text{ m/s}$ represent the density and the sound speed of air, respectively.

For the HREA, as shown in Figure 4, the air domain inside the structure can be divided into two parts. One part is the air within the embedded aperture (EA), which fluctuates with the incident sound wave and has a certain sound mass. The vibration of the air leads to viscous and heat losses, treated as acoustic resistance. The other part is the air in the chamber, which compresses or expands with the longitudinal vibration of the incident sound wave, treated as acoustic compliance.

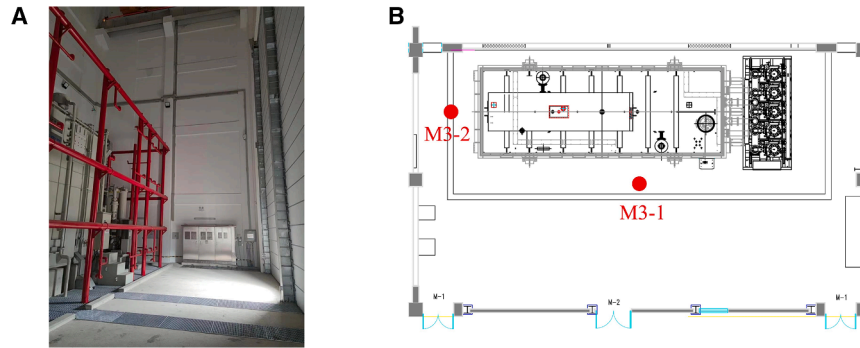


Figure 2. Field test for the transformer noise

(A) Photo of the transformer room.
(B) Acoustic MPs inside the transformer room.

The EA has length l_a and diameter d_a . The adjacent chamber has depth $L = L_u + L_b$. t is the thickness of the boundaries and b is the thickness of the EA. m and n are the dimensions of the HREA.

The impedance of the HREA is the sum of the impedances of the EA and the chamber, given by^{24–27}

$$Z = Z_a + Z_V, \quad (\text{Equation 2})$$

where Z_a and Z_V represent the impedances of the EA and the chamber, respectively.

The acoustic impedance of the perforated panel with a cylindrical EA can be derived as²⁸

$$Z_a^* = jZ_0 k(l_a + \delta), \quad (\text{Equation 3})$$

where k is the acoustic wavenumber, $\delta = [1 + (1 - 1.25\xi)] \times 4 / 3\pi d_a$ is the end correction of the acoustic mass caused by wave radiation, $\xi = 2d_a / (m + n)$.²⁹

Considering the energy dissipation caused by friction inside the EA, the acoustic impedance of the top panel featuring a cylindrical EA can be modified as³⁰

$$Z_a = Z_a^* + \theta, \quad (\text{Equation 4})$$

where $\theta = \theta_i + \theta_r$ denotes the total resistance of the EA.²⁶ θ_r and θ_i are the normalized specific radiation resistance of the EA and the corresponding EA resistance caused by internal friction, respectively.

These resistances can be described by

$$\theta_i = \frac{4R_s}{\rho_0 c_0} \frac{1}{d_a} (l_a + \delta), \quad (\text{Equation 5})$$

$$\theta_r = \frac{\pi S_a}{\lambda^2}, \quad (\text{Equation 6})$$

where $R_s = \sqrt{2\mu\rho_0\omega}/2$ is the “surface resistance,”³¹ ω is the angular frequency, and $\mu = 1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$ is the dynamic

viscosity of air. $S_a = \pi(d_a/2)^2$ is the cross-sectional area of the EA, and λ denotes the acoustic wavelength.

The impedance of the chamber (Z_V) is mainly affected by the cross-sectional area (S_c) and the volume (V). The volume of the irregular chamber, accounting for the EA and wall thickness, should be considered. The impedance of the chamber can be written as³²

$$Z_V = -jZ_0 \frac{S_c}{kV}, \quad (\text{Equation 7})$$

where $V = S_c L - S'_a l$ represents the actual volume of the chamber, with $S_c = (m - 2t)(n - 2t)$, $S'_a = \pi(d_a/2 + b)^2$, and $l = l_a - t$.

Substituting Equation 4 and Equation 7 into Equation 2 results in the following equation²⁹

$$Z = \frac{A}{S_c} Z_0 \left\{ \frac{1}{\epsilon} \left\{ \left[\frac{4R_s}{\rho_0 c_0} \frac{1}{d_a} (l_a + \delta) + \frac{\pi S_a}{\lambda^2} \right] + jk(l_a + \delta) \right\} - j \frac{S_c}{kV} \right\} \quad (\text{Equation 8})$$

where $A = m \times n$ is the cross-sectional area of the unit, $\epsilon = S_a/A$ is the perforation ratio of top panel.

Substituting Equation 8 into Equation 1 gives the final absorption coefficient of the HREA.

Based on this analysis, we designed an HR unit with an EA, inspired by the structure proposed by Huang.²⁹ The sample of the HREA is shown in Figure 5A. Since the cavity shape of the HREA unit in the literature is cylindrical, the unit cross-sectional area A in Equation 8 can be adjusted to $A' = \pi r_A^2$, and ξ can be adjusted to $\xi' = d_a/2r_A$, in which r_A is the radius of the external cylinder.

A comparison between numerical (Num.) results, simulated (Sim.) results, and experimental (Exp.) data in the reference and for the absorption coefficient of the HREA is presented in Figure 5B. Among them, the Num. result is calculated by substituting the unit dimensions in Figure 5A into Equation 8, and the Sim. result is obtained by using the finite element model established by COMSOL Multiphysics. The Experimental data are the test results in Huang et al.²⁹

From Figure 5B, it can be observed that the HREA achieves quasi-perfect absorption around 149 Hz (reaching 1 in the experiment). The numerical and simulated results of the resonance frequency (f_r) and the absorption peak (α_{peak}) for the HREA are greatly consistent with the experimental data, which

Table 1. Noise levels of the converter transformer (dB)

Frequency band	20–500 Hz	20–1,000 Hz	20–6,300 Hz
MP1	90.08	90.09	90.09
MP2	89.14	89.16	89.16
MP3	90.33	90.33	90.34
MP4	89.47	89.49	89.49

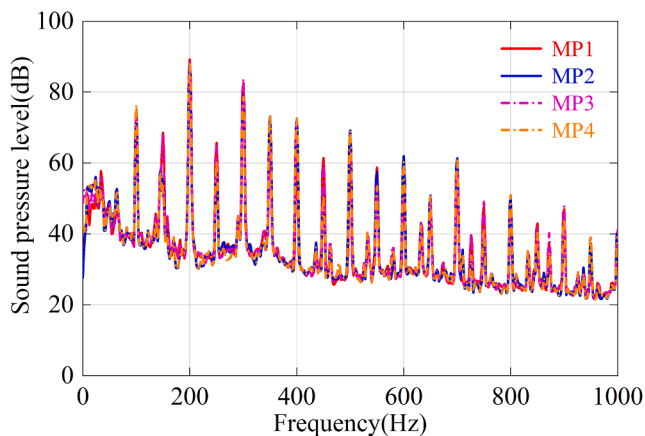


Figure 3. Narrowband spectral features of SPLs for the transformer

demonstrates the effectiveness of the HREA's structural design and the accuracy of the numerical and simulation models.

Design of sound-absorbing AM

From Equation 8, the absorption performance of the HREA is primarily influenced by parameters such as the diameter (d_a) and the length (l_a) of the EA, the depth (L) of the adjacent chamber, and the dimensions of the HREA's edges (m and n). By adjusting these parameters, absorption coefficient curves with different absorption peaks and bandwidths can be obtained.

To effectively control transformer noise, and taking into account the size limitations of the noise barrier on the AM, HREA

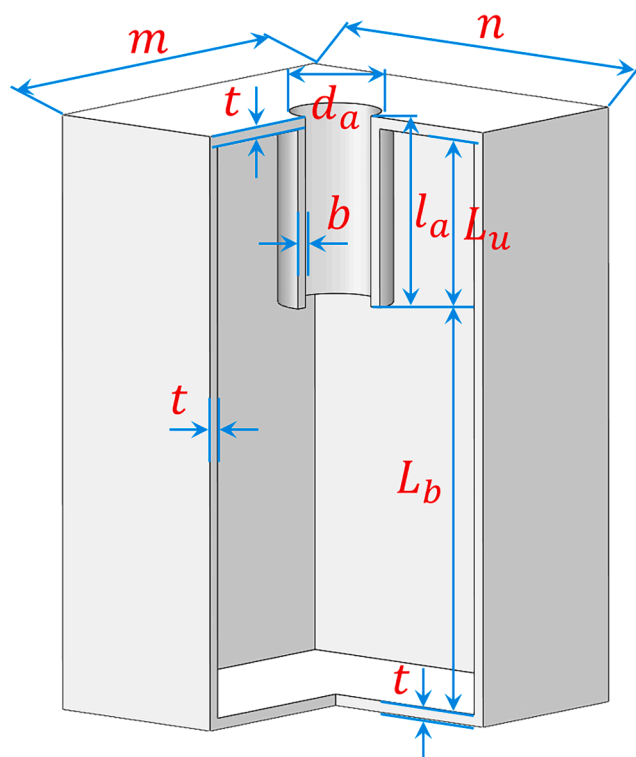


Figure 4. Internal structure diagram of the HREA

units with higher absorption coefficients and wider absorption bandwidths were selected. Each HREA unit was designed with a height of 44 mm, a top, bottom, and side plate thickness of 3 mm, an EA thickness of 1 mm, and a chamber depth of 38 mm. By adjusting the diameter and length of the EA, HREA units can achieve ideal absorption peaks and absorption bandwidths at target frequencies. The structural parameters of each HREA unit are listed in Table 2.

In order to effectively control the transformer noise at multiple peak frequencies, an AM with multiple absorption peaks was created by combining the above HREA units in parallel, as shown in Figure 6. The AM is 206 mm long, 104 mm wide, and 44 mm high, with the top, bottom, and side plates having a thickness of 3 mm and the EA thickness being 1 mm.

Absorption performance of the AM

To evaluate the absorption performance of the HREA units and the AM, we developed acoustic finite element method (FEM) models for both individual HREA units and the AM using COMSOL Multiphysics with the acoustic-thermoacoustic interaction module. Each model consisted of three main components: the perfect matched layer acoustic domain, the incident acoustic domain, and the thermoviscous acoustic domain. The walls of the model were assumed to be acoustically hard boundaries, as illustrated in Figure 7A.

The air domain was defined as a pressure acoustic domain with a thermoviscous impedance boundary. An incident port was added to the upper end surface of the incident acoustic domain. The incident wave, with an excitation pressure of 1 Pa, was directed vertically into the AM. The grid for the pressure acoustic frequency domain analysis was controlled using COMSOL's built-in physical fields, as shown in Figure 7B. The complete grid contained 64,493 domain elements, 14,624 boundary elements, and 2,186 edge elements, resulting in a total of 1,348,486 degrees of freedom.

Figure 8 shows the acoustic characteristics of HREA units and the AM simulated through FEM, including the acoustic surface impedance and absorption coefficient. Z_s represents the normal surface impedance of the structure, which can be calculated from the ratio of the sound pressure (p) to the particle velocity (v) at the upper surface. Absorption coefficient is a key parameter for evaluating the absorption performance of a material or structure and can be calculated using Equation 1.

The calculation results in Figure 8A show that, HREA units did not achieve perfect absorption at their resonance frequencies, the absorption coefficient gradually decreased with increasing resonance frequency. And there were deviations between resonance frequencies and target frequencies, with most resonance frequencies being higher than target frequencies. Fortunately, the AM, which was composed of multiple HREA units, achieved superior absorption performance ($\alpha \geq 0.8$) at each target frequency (100 Hz and its harmonics). Combining this with the normalized surface impedance in Figure 8B, it can be seen that for HREA units, the imaginary part of their normalized surface impedance did not reach zero near 100 Hz and its harmonics, unlike the AM, leading to an error between actual resonance frequencies and target frequencies. Secondly, although the acoustic surface impedance of the AM deviated significantly from acoustic impedance matching, its acoustic resistance

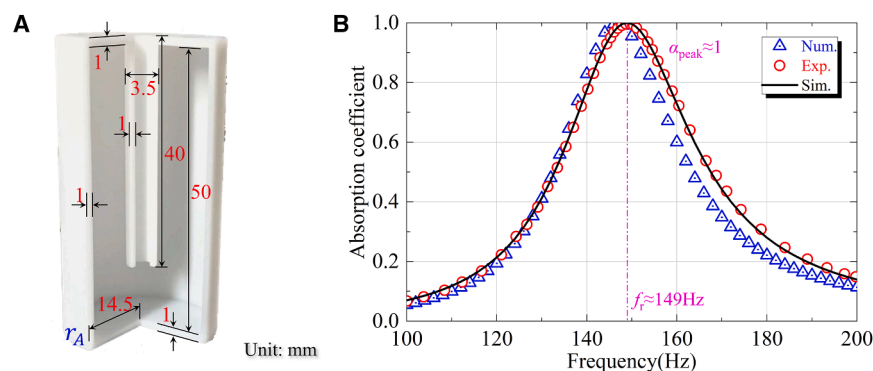


Figure 5. Internal structure and absorption performance of an HREA

(A) Dimensions of the referred unit.

(B) Exp., Num., and Sim. absorption coefficients.

component was significantly larger than those of HREA units, which could effectively suppress the amplitude of the reflection coefficient, thus achieving better absorption performance. In contrast, the acoustic resistance components of HREA units were too small, resulting in a serious mismatch of acoustic impedance and significant sound reflection.

To clarify the reasons behind the shift in absorption peaks of the AM with multiple HREA units combined in parallel, and the improvement in absorption performance at target frequencies, further analyses were conducted here based on particle velocity and energy dissipation. First, Figure 9 presents the simulated flow fields of particle velocity vectors near the necks of the AM at target frequencies, for further analysis of the coherent coupling effect between units. From Figure 9, notable coupled flow patterns between units of the AM are observed with opposite vector directions in the adjacent necks.

Taking 100 Hz as an example (Figure 9A), the neck of the unit with a target frequency of 100 Hz (HR100 Hz) had opposite vector directions to those of necks of adjacent units (target frequencies of 200/300/400/500 Hz), and velocity vectors were parallel to the surface and point from necks of adjacent units to the neck of HR100 Hz. Due to the coherent coupling effect, the air particles in the AM only vibrate laterally.³³

Similarly, at 200 Hz (Figure 9B), coupled flow patterns can be observed between the unit with a target frequency of 200 Hz (HR200 Hz) and adjacent units (target frequencies of 300/400/500 Hz). At 300 Hz (Figure 9C), coupled flow patterns appear between the unit with a target frequency of 300 Hz (HR300 Hz) and the adjacent unit (target frequency of 200 Hz). At 400 Hz (Figure 9D), coupled flow patterns appear between the unit with a target frequency of 400 Hz (HR400 Hz) and adjacent units (target frequencies of 200/300 Hz). At 500 Hz (Figure 9E), coupled flow patterns appear between the unit with a target frequency of

500 Hz (HR500 Hz) and adjacent units (target frequencies of 200/300/400 Hz). The coherent coupling effect between HREA units was the main reason for the shift between absorption peaks of the AM and resonance frequencies of HREA units.

For the HREA unit, the acoustic energy dissipation within the EA accounted for the majority of the total energy dissipation of the unit.³⁴ Therefore, comparative analyses of the EA energy dissipation of the HREA unit and the AM were conducted here. Figure 10 visually shows the acoustic energy dissipation within the EA of HREA units and the AM. It can be seen that for the AM, the energy dissipation within EAs showed significant peaks at 100 Hz and its harmonics, which corresponded exactly to peak frequencies of the absorption coefficient of the AM in Figure 8A, and the energy dissipation peaks were basically consistent. However, for HREA units, as the resonance frequency of each unit increased, the acoustic energy dissipation within the EA decreased. The energy dissipation peaks of the AM were far higher than those of HREA units. This is the main reason why the absorption performance of the AM at target frequencies was superior to that of HREA units.

In addition, the effect of sound-wave incidence angle on the absorption performance of the AM was investigated. Figure 11 intuitively shows the simulated absorption coefficients of the AM at different sound-wave incidence angles, including narrow-band and 1/3 octave band results.

From Figure 11A, it can be seen that changes in the sound-wave incidence angle caused shifts in the peak frequency of absorption coefficient for the AM, and the shift became more pronounced with increasing incidence angle. Furthermore, as the sound-wave incidence angle increased, the absorption peak of the AM generally increased first and then decreased, with the highest absorption peak mostly occurring at an incidence angle of 40°–60°. For frequency bands outside the peak frequency, the absorption performance of the AM also gradually improved with the increasing sound-wave incidence angle.

Figure 11B compares the measured data and simulated result of the absorption coefficient for the AM in the 1/3 octave band, and also provides the simulated absorption coefficient of the AM in the 1/3 octave band at different sound-wave incidence angles. The oblique-incidence absorption coefficient α_θ of the AM was simulated through FEM, and the absorption coefficient in the 1/3 octave band was calculated from the absorption coefficient at narrowband frequencies based on the energy averaging principle. The random-incidence absorption coefficient α_r of the AM was measured in a reverberation chamber based on the ISO 354 standard. The equivalent random-incidence absorption coefficient α_{diff} of the AM was derived through the weighted

Table 2. Geometric parameters of HREA units (mm)

Parameter	<i>m</i>	<i>n</i>	<i>L</i>	<i>d_a</i>	<i>l_a</i>
100 Hz	130	98	38	8.4	24.5
200 Hz	67	48	38	8.4	24.5
300 Hz	32	47	38	8.4	23.4
400 Hz	32	26	38	8.4	24.5
500 Hz	32	18	38	8.4	22.1

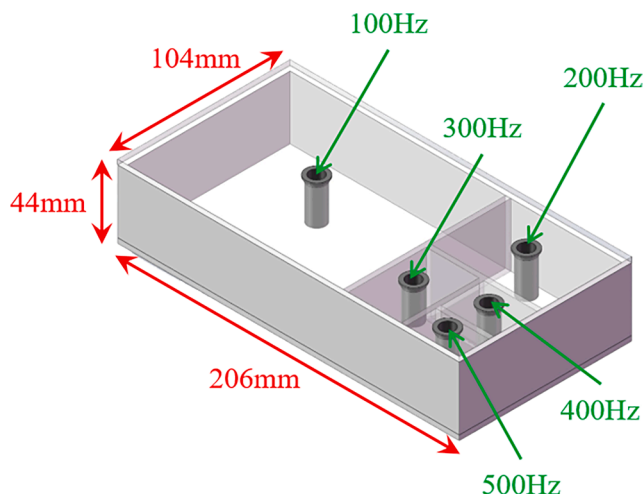


Figure 6. The AM consisted of HREA units in parallel

average of the oblique-incidence absorption coefficients α_θ at different incidence angles.

Analysis of Figure 11B reveals that as the sound-wave incidence angle increased, the oblique-incidence absorption coefficient α_θ of the AM increased within the 1/3 octave band, with the normal-incidence absorption coefficient α_0 under perpendicular wave incidence showing the lowest absorption level. Furthermore, the equivalent random-incidence absorption coefficient α_{diff} of the AM generally fell below the measured random-incidence absorption coefficient α_r across most frequency bands.

The main reasons for the above discrepancies are that the RCM yielded a statistical random-incidence absorption coefficient, where the sound field was assumed to be a fully diffused sound field, and the absorption coefficient was the statistical average of all incidence angles, polarization directions, and spatial modes. However, the result simulated through FEM model was the normal-incidence absorption coefficient, which was an ideal and controllable single-angle absorption coefficient. The FEM model cannot reproduce the grazing-incidence sound wave, which was highly prevalent in the RCM. And additional scattering and dissipation will occur at the edges of samples in the reverberation chamber, resulting in an increase in equivalent sound absorption. In addition, arrays of multiple AM

were required for the RCM, and each AM had manufacturing and installation errors. Furthermore, the coherent coupling effect existed between each HREA unit and each AM, resulting in a deviation between the resonance frequency of each HREA and the design frequency. After stacking multiple AMs, a group of resonance peaks was formed within the 1/3 octave band instead of isolated peaks. These are main reasons why the absorption coefficient of the AM tested under random-incidence sound waves in the RCM was usually higher than that calculated under normal-incidence or oblique-incidence sound waves here.^{35,36}

Broadband noise reduction barrier

In this study, the design of the noise barrier was guided by universal design principles to effectively control transformer noise across a range of acoustic characteristics.^{37,38} Drawing on the sound absorption mechanism of Helmholtz resonance and the universal design principles, the following concepts were implemented to enhance the absorption performance and broaden the noise reduction bandwidth in the mid-low frequency band.

- 1 Material selection: the noise barrier was primarily constructed from aluminum alloy (AL), porous cement-based silicone rubber materials,³⁹ aluminum foam (AF), and acrylonitrile butadiene styrene (ABS) materials. This combination not only enhances absorption and insulation performance but also reduces maintenance needs (following the principle of low physical effort).
- 2 Composite sound-absorbing system: the barrier's composite sound-absorbing system includes PRM, PMM, and AMs. This combination optimizes absorption performance across different frequency bands. The metal components provide sound insulation and protect the acoustic materials from being damaged (in line with the principles of equitable use and tolerance for error).
- 3 Modular design for flexibility: the chamber and EA of the HREA unit was produced separately. By assembling chambers and EAs of various sizes, HREA units with different resonance frequencies were achieved. This modular design allows the resonance frequency of the AM to be adjusted based on the dimensions of the chamber and EA (aligned with the principles of flexibility in use and perceptible information).
- 4 Ease of maintenance: the design incorporates adequate size and space within the noise barrier panel, facilitating easy disassembly and maintenance by operators

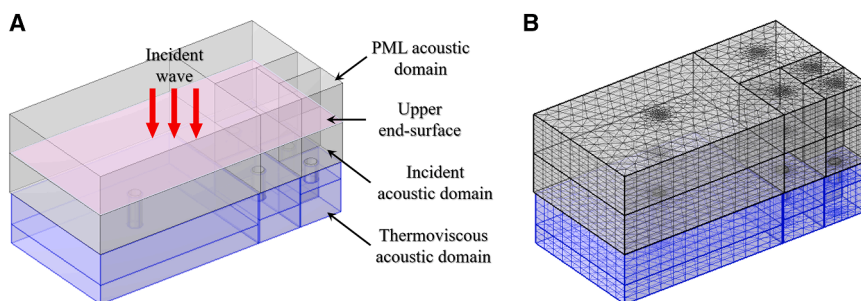


Figure 7. FEM model used to simulate the absorption performance of the AM

(A) Compositions of the acoustic FEM model.
(B) Grid partition of the AM.

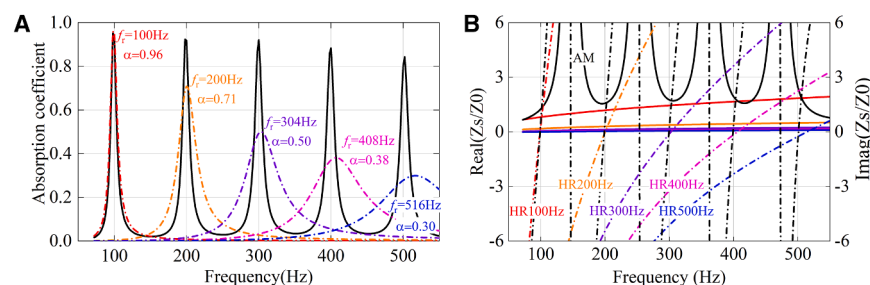


Figure 8. Simulated acoustic characteristics of HREA units and the AM at narrow-band frequencies

(A) Absorption coefficients.
(B) Normalized surface impedance.

(following the principles of simple and intuitive use, and size and space for approach and use).

- Based on these principles, a broadband noise reduction barrier was designed and assembled, following flowchart shown in Figure 12. The actual prototype of the barrier is also depicted in the figure.

RESULTS

Reverberation chamber test

In accordance with ISO 354 standard, we measured the absorption performance of the new barrier panel through the RCM. Absorption coefficients of the new barrier panel were compared to those of two existing panels, as shown in Figure 13. Panel #1 consisted of a 1.5 mm thick AL fiber plate with a 120 mm air back cavity,⁴⁰ while panel #2 was made of a micro-perforated plate, an AL fiber plate, and a 120 mm air back cavity.⁴¹ Figure 13 shows the spectral features of the absorption coefficients for these structures.

Existing noise barrier panels exhibited limitations in absorption performance, particularly in the mid-low frequency band, where

the absorption bandwidth was narrow.^{42,43} To address this, it is necessary to incorporate a sufficiently thick air back cavity.^{24,44,45} The new barrier panel, combining the PRM, PMM, and the low-frequency sound-absorbing AMs, overcame these limitations by increasing the absorption coefficient and absorption bandwidth in the mid-low frequency band. The average absorption coefficient of the new barrier panel in the frequency band below 500 Hz reached 0.80, with a significant absorption coefficient of 0.43 at 100 Hz.

We established a local FEM model of the barrier panel to calculate the absorption coefficient of the panel, and to investigate the influence of internal components on its absorption performance. The results show that the AM notably enhanced the absorption performance in the low-frequency band, especially below 400 Hz, while the PRM improved the absorption performance in the mid-high frequency band (above 500 Hz).

Furthermore, using the finite element model of the barrier panel, we adjusted and optimized the internal components of the new barrier panel in an attempt to achieve improved absorption performance, or reduce its overall thickness.

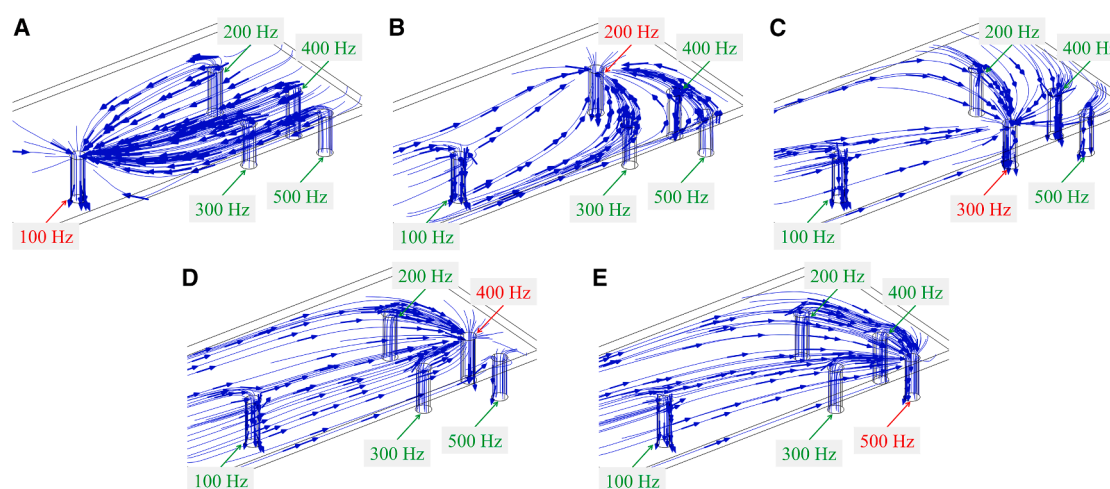


Figure 9. Particle velocity vector fields near EAs of the AM at peak frequencies

- 100 Hz.
- 200 Hz.
- 300 Hz.
- 400 Hz.
- 500 Hz.

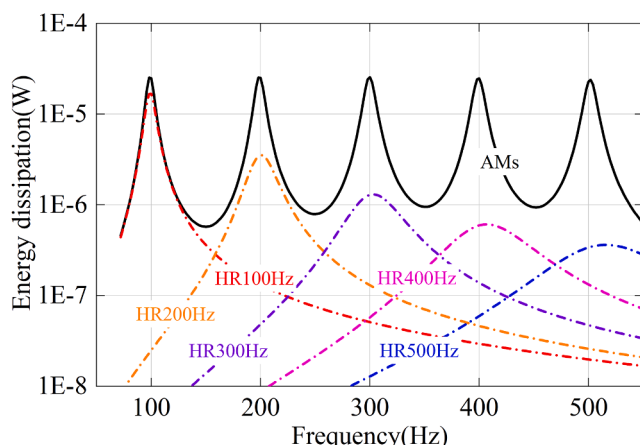


Figure 10. Acoustic energy dissipation of the EA for HREA units and the AM

Pilot application and NRE analysis

To assess the practical noise reduction performance of the new barrier panel, a pilot application and field tests were conducted. Due to site constraints and safety considerations, we selected the converter transformer in the ZTD station for test. Noise levels outside the transformer room were measured before and after installing the barrier. The size of each barrier panel is 1,050 mm × 500 mm, and the thickness is 140 mm. 24 panels are assembled by five columns (H beam) and four beams (steel plates) to form a noise barrier structure with a width of 4 m and a height of 3 m, as shown in Figure 14A. Bases are installed at the four column feet of the barrier structure to maintain its stability.

The new barrier was placed 3 m from the room door, as depicted in Figure 14B. According to the ISO 10847:1997 standard, the reference point M0 was placed 1.5 m above the top of the barrier to monitor the sound source equivalence. M1–M4 were located outside the barrier at distances of 1, 2, 4, and 8 m, respectively, at a height of 1.5 m. M5 and M6 were positioned 1 m outside and inside the barrier, respectively, to assess barrier's impact on the sound field within the barrier. The temperature at the test site was 32°C, wind speed was 2.6 m/s, humidity was 75%, and atmospheric pressure was 0.94 bar.

Based on ISO 10847:1997 standard, the SPL at each MP before and after the installation of the new noise barrier was

directly tested under the same test environment and operating power of the transformer. Noise data at each MP before and after the installation of the new barrier were collected three times, with each collection lasting for 5 min. The average value of the equivalent continuous SPL ($L_{peq,T}$) from three sets of data at each MP was used as the noise level evaluation index. The ILs of the new noise barrier was calculated using Equation 9, and the NREs of the new noise barrier were analyzed based on the total IL value and its spectral characteristics.

$$D_{IL} = (L_{ref,A} - L_{ref,B}) - (L_{r,A} - L_{r,B}), \quad (\text{Equation 9})$$

where $L_{ref,A}$ and $L_{ref,B}$ are SPLs at the reference point after and before installing the barrier. $L_{r,A}$ and $L_{r,B}$ are SPLs at MPs after and before installing the barrier.

(A) The new noise barrier used in field tests. (B) Photo of field tests and MPs.

Table 3 summarizes the ILs of the new noise barrier compared to a rigid barrier (non-absorptive) and two existing absorbing barriers at different MPs outside the transformer room.

The results are based on a simulation model. The rigid barrier reduced the noise levels by 5.3–8.4 dB at M1–M4, and two absorbing barriers further reduced the noise by 0.3–3.2 dB. The new noise barrier provided additional NREs of 0.4–2.2 dB compared to two absorbing barriers.

To better assess the noise reduction performance of the new barrier, we analyzed the additional NRE at M1 compared to the rigid barrier, as M1 is closer to the barrier and less influenced by diffracted sound from sides of the barrier. Table 4 provides the additional NRE at M1 for various barriers in the 1/3 octave band, and Figure 15 shows the narrowband spectral features of the additional NRE achieved by the new noise barrier and two absorbing barriers compared to the rigid barrier.

Table 4 reveals that the new barrier and two absorbing barriers achieved additional NREs at each 1/3 octave band within 500 Hz compared to the rigid barrier. Notably, the new barrier achieved additional NREs of over 4 dB in the 1/3 octave band of (250, 400) Hz compared to the rigid barrier, which were far superior to those offered by the traditional absorbing barriers.

From a narrowband spectrum perspective, Figure 15 reveals that the new barrier effectively enhanced the additional NRE compared to traditional absorbing barriers across most frequencies. Furthermore, the additional NREs of each barrier at 100 Hz and its harmonics are also clearly indicated in Figure 15. In addition, it can be observed that at few frequencies,

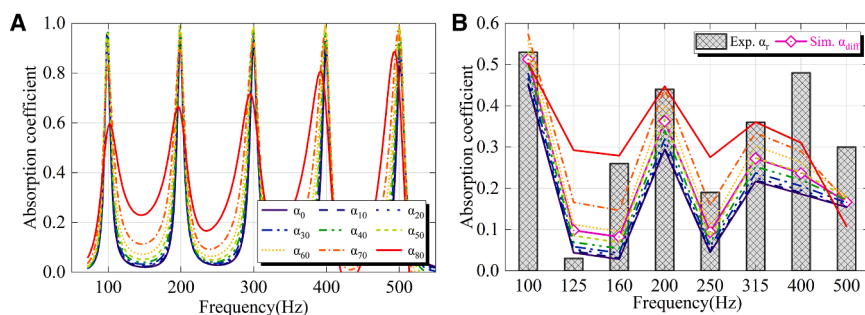


Figure 11. Absorption coefficient of the AM at different sound-wave incidence conditions

(A) At narrowband frequencies.
(B) In 1/3 octave bands.

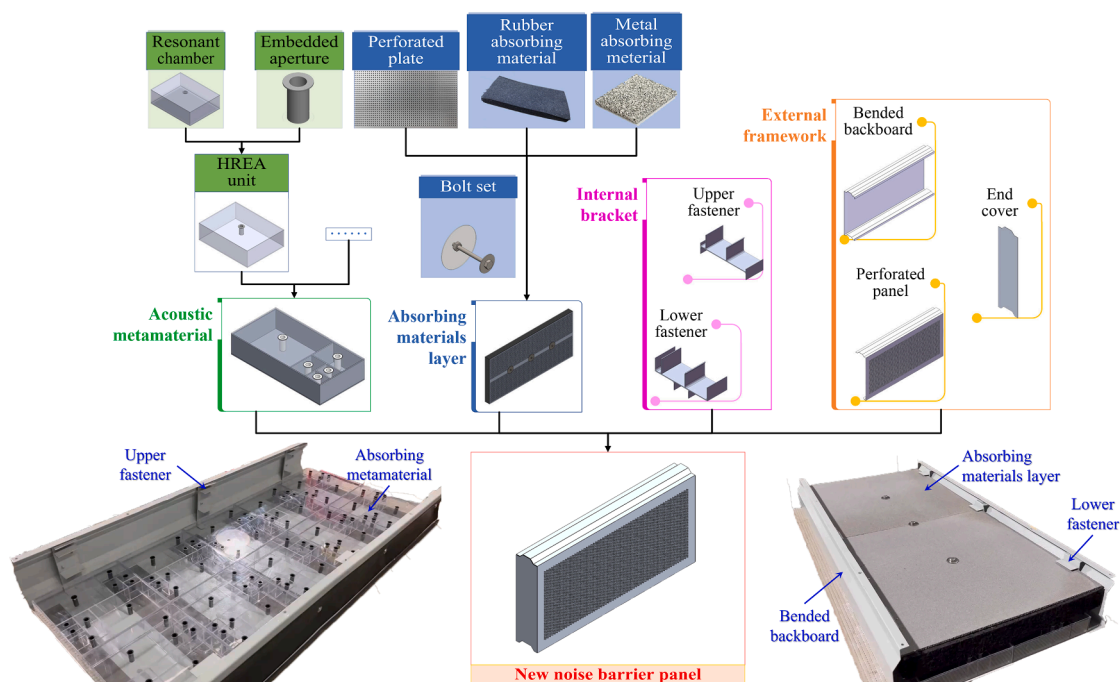


Figure 12. Design, assembly flowchart, and prototype of the new barrier panel

absorbing barriers and the new barrier compared to the rigid barrier were negative. This phenomenon is likely caused by sound wave interference and reflection effects in the test environment, which is a common occurrence in noise barrier applications.^{46,47}

Similarly, we utilized the acoustic BEM prediction model for outdoor noise barriers to simulate the NRE of the optimized new barrier. In summary, by adjusting the internal components of the new barrier panel, such as optimizing the HREA units within the AM and providing a micro-perforated structure on its surface, the NRE of the new barrier can be further improved by

0.4–1.1 dB. In addition, by adjusting the thickness of the internal components of the panel, such as eliminating the air cavity between the perforated panel and the PMM, and appropriately reducing the thickness of the PRM and the air cavity between the perforated plate and the AMs, the overall thickness of the new barrier panel can be reduced to 100 mm while maintaining an NRE comparable to the original new barrier.

DISCUSSION

To mitigate the impact of transformer noise on the environment, researchers conducted field tests on a typical transformer in China. Based on the absorption mechanism of Helmholtz resonance, a high-performance HREA unit for low-frequency sound absorption was designed. Using the coherent coupling effect, multiple weak-resonance HREA units were combined in parallel to obtain an AM with multiple target absorption peaks. Combining PMM and PRM sound-absorbing materials, and following general design principles, a broadband noise reduction barrier panel was developed. The new barrier panel was then adjusted and optimized to improve its NRE or reduce its overall thickness. Through test, simulation, and analysis, the following conclusions were drawn.

- 1 The transformer noise primarily concentrated in the mid-low frequency band of (20, 500) Hz, with several SPL peaks observed at 100 Hz and its harmonics, indicating discrete frequency noise components.
- 2 Utilizing the coherent coupling effect, multiple weakly resonant HREA units with low absorption peaks can be combined in parallel to obtain an AM that achieves excellent

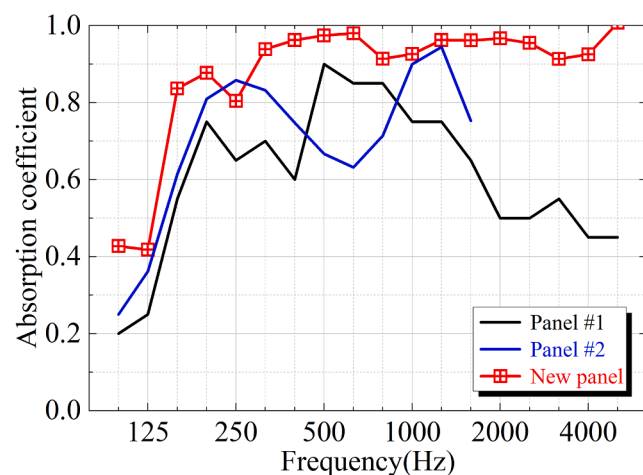


Figure 13. Absorption coefficients of panels in 1/3 octave bands

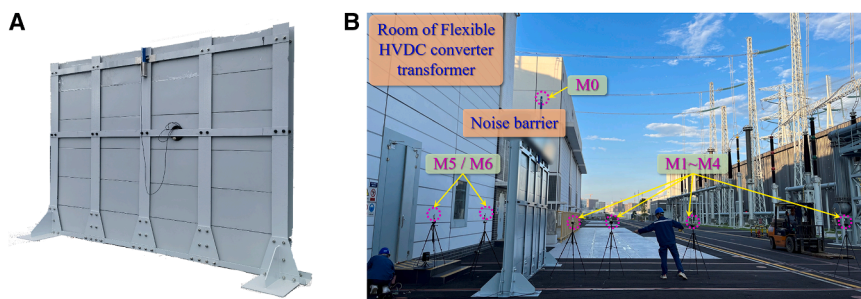


Figure 14. Pilot application and field test of the new noise barrier

(A) The new noise barrier used in field tests.
(B) Photo of field tests and MPs.

absorption performance ($\alpha \geq 0.8$) at 100 Hz and its harmonics.

- 3 The novel barrier panel combined PRM, PMM, and AMs, achieving a measured average absorption coefficient of 0.80 in the frequency band of (100, 500) Hz. Notably, the absorption coefficients in the frequency band of 200/315 Hz, where outdoor noise from transformers was significant, reached 0.88/0.94, respectively.
- 4 Installing a rigid barrier 4 m outside the transformer room can achieve an IL of 8.4 dB, and existing absorbing barriers can further increase the IL by 2.3/3.2 dB. Furthermore, the new barrier can additionally reduce noise by 1.3/2.2 dB compared to existing absorbing barriers, with a significant improvement in NRE, especially in the frequency band of (200, 400) Hz.
- 5 By adjusting the HREA units within the AM and introducing micro-perforated structures on its surface, the absorption performance of the new barrier panel can be further improved, resulting in an increase of 1.0 dB in IL. Based on the optimized new barrier panel, appropriately reducing the thickness of internal absorbing components and air cavity can reduce the overall thickness of the panel to 100 mm, while maintaining an NRE comparable to that of the unoptimized new panel.

Limitations of the study

This study has several limitations. First, the simulation model for the absorption performance is mainly based on idealized incidence conditions, whereas the reverberation chamber measurements involve a diffuse sound field with random incidence. This difference in acoustic field assumptions may lead to discrepancies between the simulated and experimental absorption coefficients of the metamaterial-based panel, especially near resonance frequencies. Second, the present work focuses primarily on sound absorption and noise reduction performance, and

does not provide a systematic comparison of sound insulation performance with conventional noise barriers or composite panels having the same surface density. Such mass-matched comparisons would further strengthen the reliability of the conclusions regarding the engineering advantages of the proposed metamaterial-based barrier. In addition, the validation is conducted for a representative substation scenario, and the performance may vary under different source characteristics and installation conditions, which should be investigated in future studies.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Xing Du (duxing@swjtu.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The data supporting the findings of this study are available within the manuscript and the supplemental information.
- This study does not generate original code.
- Any additional information required is available upon reasonable request to the [lead contact](#).

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AUTHOR CONTRIBUTIONS

Conceptualization, S.X. and L.L.; methodology, S.X., and X.X.; investigation, S.X., L.L., Q.W., and W.M.; writing – original draft, S.X. and L.L.; writing – review and editing, Q.W., W.M., X.D., X.X., and Z.W.; funding acquisition, L.L.

Table 3. ILs at MPs after installing various noise barriers (dB)

Measurement point	M1	M2	M3	M4
Rigid barrier (Sim.)	8.4	8.4	5.3	7.4
Absorbing barrier #1 (Sim.)	10.7	9.9	6.8	7.7
Absorbing barrier #2 (Sim.)	11.6	10.4	7.1	7.8
New barrier (Sim.)	12.9	11.3	7.5	7.4

Table 4. Additional NREs at M1 in the 1/3 octave bands compared to the rigid barrier (dB)

1/3 octave band (Hz)	100	125	160	200	250	315	400	500
Absorbing barrier #1	0.4	1.3	0.4	1.2	2.5	2.7	1.5	1.4
Absorbing barrier #2	0.5	1.7	0.5	1.4	3.9	3.7	2.3	1.1
The new barrier	1	2.4	0.6	2.1	4.8	5.6	4.7	1.9

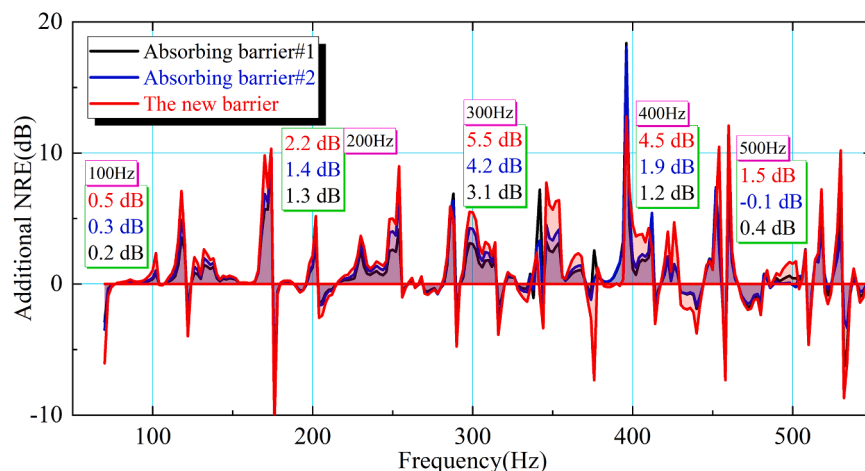


Figure 15. Narrowband spectral features of additional NREs at M1 compared to the rigid barrier

and X.X.; resources, L.L., X.D., X.X., and Z.W.; supervision, L.L., X.D., X.X., and Z.W.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the writing of this article, the authors did not use generative AI or AI-assisted technologies.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
PULSE Reflex	Brüel & Kjaer	https://www.hbkworld.com/zh/services-support
SolidWorks	Dassault Systèmes SolidWorks Corporation	https://www.solidworks.com/sw/support/downloads.htm
Origin	OriginLab Corporation	https://www.originlab.com/demodownload.aspx
Microsoft PowerPoint	Microsoft Corporation	https://www.microsoft.com/zh-cn/microsoft-365
COMSOL Multiphysics	COMSOL China Co., LTD	https://cn.comsol.com/
LMS Virtual. Lab Acoustics	SIEMENS	https://www.siemens.com/en-us/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study does not use experimental model and subject details typical in the life sciences.

METHOD DETAILS

Sound absorption test for AM samples

The random-incidence absorption coefficient of the acoustic metamaterial (AM) samples was measured according to ISO 354 using a Mecanum Acoustic Test Cabin (MATC; Figure S1A). The MATC operates over 100–10,000 Hz with a maximum sound pressure level of 110 dB and is equipped with FEA-optimized sound deflectors and four uncorrelated loudspeakers to generate a diffuse sound field (Figure S1B). Each AM specimen measured $206 \times 206 \times 45$ mm and consisted of two AM plates (Figure S2A). Measurements were conducted at 20°C, 50% relative humidity, and 1.0 bar atmospheric pressure.

Two configurations, containing 9 and 16 samples with total exposed areas of 0.38 m² and 0.68 m², respectively, were tested (Figures S2B and S2C). For each configuration, measurements were performed at three sample positions with three repeats per position. The nine measurements were averaged for each configuration, and the reported absorption coefficient was taken as the mean of the 9-sample and 16-sample datasets. The resulting 1/3-octave-band data are listed in Table S1.

Configuration of the new barrier panel

The barrier panel comprised a perforated front panel, a perforated inner plate, a porous rubber material (PRM) layer, a porous metal material (PMM) layer, AM units, fasteners, end covers, a bent backboard, and bolt sets (Figure S3A). The PMM, PRM, and perforated plate formed the upper absorbing layer, whereas the AM units were bonded to the bent backboard (Figure S3B). The external dimensions of the assembled panel were $1050 \times 500 \times 140$ mm.

Detailed component information is provided in Table S2. The perforated front panel and inner perforated plate were made of 6063 aluminum alloy, with perforation diameters/perforation ratios of 6 mm/32.1% and 4 mm/22.4%, respectively. The PMM was a closed-cell aluminum foam with a density of 350 kg/m³. Ten ABS AM units ($206 \times 206 \times 40$ mm each) were installed in the lower panel cavity. A 19 mm air cavity was retained between the perforated front panel and PMM, and a 23 mm air cavity was retained between the perforated plate and AM layer.

Finite-element simulation and optimization of panel absorption

Panel absorption was simulated in COMSOL Multiphysics using the Pressure Acoustics module. To reduce computational cost while preserving periodicity, a representative sub-model covering the area of two AM units was used ($206 \times 206 \times 140$ mm), with periodic boundary conditions applied to the four lateral sides. The model included a perfectly matched layer, an incident acoustic domain, and the panel subdomains corresponding to the perforated panel, upper air cavity, PMM, PRM, perforated plate, lower air cavity, and AM layer (Figure S4A). A normally incident plane wave with an amplitude of 1.0 Pa was imposed at the incident port.

The PMM and PRM were described using the Johnson–Champoux–Allard model, with material parameters listed in Table S3. Narrow-region acoustic features were assigned to the perforated panel, perforated plate, and HREA necks to capture thermoviscous losses. The mesh was generated using COMSOL's physics-controlled settings and contained 1,430,425 total degrees of freedom (Figure S4B). The model was used to calculate absorption coefficients up to 3150 Hz and to compare the full panel with configurations excluding the PMM, PRM, perforated plate, or AM layer (Figure S5).

The AM layer was then optimized by enlarging the cross-sectional areas of the 200 and 300 Hz HREA units while preserving the original footprint; the optimized dimensions are given in Table S4. Micro-perforated structures were further introduced on the AM

surface outside the embedded-neck region, with a perforation diameter of 0.1 mm and a perforation ratio of 0.2%, to lower off-resonance surface impedance and improve broadband sound penetration (Figures S6 and S7). A thinner optimized panel was also examined by removing the upper air cavity, reducing the PRM thickness to 40 mm, and setting the cavity between the perforated plate and AM to 12 mm; the resulting absorption coefficients are shown in Figure S8.

Reverberation-room measurement of barrier-panel absorption

The random-incidence absorption coefficient of the full barrier panel was measured in a reverberation room according to ISO 354 (Figure S9). The room dimensions were $7.5 \times 6.5 \times 4.2$ m (209 m^3). Twelve omnidirectional sound sources were placed in three corners, and six microphones were randomly distributed within the chamber. The sample was installed off-center and with its edges non-parallel to the room walls, and non-absorbing fixtures were used to seal the sample edges and suppress lateral absorption (Figure S10A). For the empty-room measurement, both the sample and fixtures were removed while all other conditions were kept unchanged (Figure S10B).

The exposed sample size was 3.68×2.90 m (9.50 m^2). The panel thickness was 140 mm and the surface density was 34.5 kg/m^3 . Measurements were performed at 26°C , 85% relative humidity, and 0.94 bar atmospheric pressure. The sound source was moved to three positions, and six microphone responses were recorded at each position. The final absorption coefficient was obtained by averaging 18 datasets. Reverberation times measured for the empty-room and sample-installed conditions, together with the derived equivalent absorption area and absorption coefficient, are summarized in Table S5.

Boundary-element simulation of outdoor noise reduction effect

The outdoor noise reduction effect of the barrier was evaluated using an acoustic boundary element method (BEM), which accounts for structural geometry, surface absorption, and sound-wave diffraction and reflection. Air density and sound speed were set to 1.21 kg/m^3 and 344 m/s, respectively. For the 10–1000 Hz calculation range, the element length was set to 56 mm to satisfy the requirement that the mesh size remain smaller than one-sixth of the shortest wavelength. The primary source term was defined using the measured sound pressure level at 1 m outside the door of the flexible DC transformer, and additional sources representing the phase A and phase B transformers of Unit 2 were included to reproduce the multi-source operating condition.

Because the BEM requires acoustic impedance rather than absorption coefficient as input, the measured absorption data were converted to equivalent impedance using the conversion formulas between the absorption coefficient and the acoustic impedance presented by Delany and Attenborough.^{48,49} The reverberation-room absorption coefficient was approximated as the normal-incidence absorption coefficient, and the impedance was then fitted using the Attenborough dual-parameter model. The impedance boundary was mainly assigned to the inner surface of the barrier screen. The geometric model of the transformer room and barrier is shown in Figure S11. The transformer room dimensions were $57 \text{ m} \times 11.6 \text{ m} \times 21 \text{ m}$, and the ground and room walls were treated as rigid reflective boundaries. A reference point (M0) and evaluation points (M1–M6) were arranged outside the transformer room according to the field-test layout (Figure S12).

Model validation was performed by comparing simulated and measured outdoor sound pressure levels for the cases without a barrier and with the new barrier installed. The results are listed in Tables S6 and S7. The maximum deviation was 3.3 dB and the maximum relative error was 4.6%, supporting the use of this model for engineering evaluation of barrier noise reduction performance.

QUANTIFICATION AND STATISTICAL ANALYSIS

The transformer noise and outdoor sound pressure levels after installing the new noise barriers in this paper were obtained from original data collected by the Brüel & Kjaer testing system and then processed by PULSE Reflex. The charts were generated from the original data using Origin and Microsoft PowerPoint. The absorption performance of the AM and the new barrier panels was calculated using COMSOL Multiphysics. The outdoor sound pressure levels after installing traditional noise barriers and the optimized new barriers were simulated using LMS Virtual. Lab Acoustics.